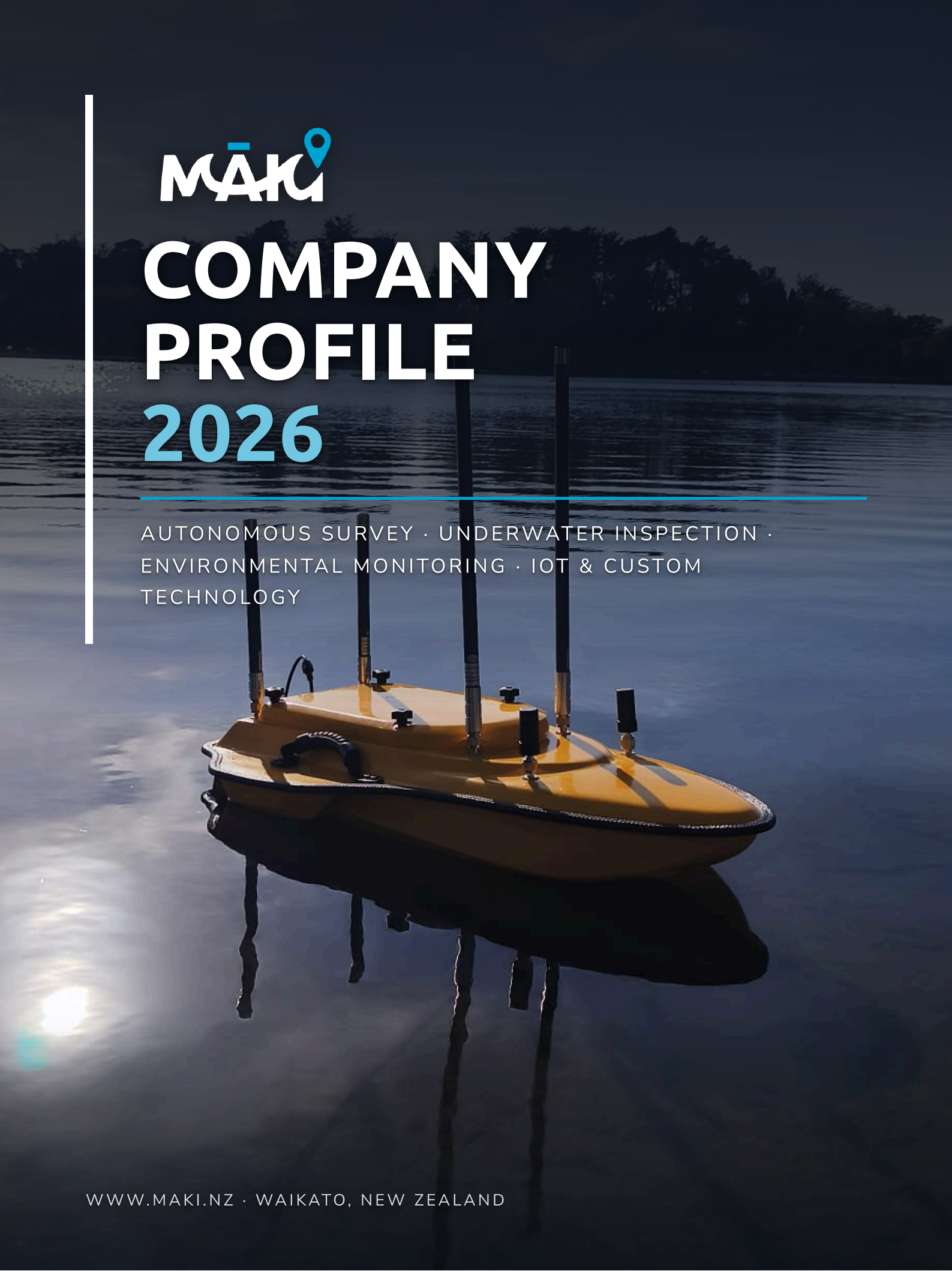


COMPANY PROFILE 2026

AUTONOMOUS SURVEY · UNDERWATER INSPECTION · ENVIRONMENTAL MONITORING · IOT & CUSTOM TECHNOLOGY

WWW.MAKI.NZ · WAIKATO, NEW ZEALAND

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ABOUT MĀKI

We go where it's hard to measure

MĀKI is a New Zealand engineering technology company, founded in 2020 and based in the Waikato.

Our team of six designs, builds and operates autonomous systems (uncrewed survey boats, underwater ROVs, ground rovers, drones and IoT sensor networks) and turns the data they collect into answers for councils, utilities, contractors and researchers.

Most companies are strong in hardware *or* software. We do both in-house: mechanical, electronic and communications engineering alongside our own control software, data platforms and AI. That's why we can build the right tool for a job rather than make the job fit the tool.

Our mission is to make cutting-edge engineering technology accessible to the organisations that manage New Zealand's natural and built environments. That means collecting data safely and efficiently, including in places people shouldn't have to go.

MĀKI at a glance

Since 2020 we've surveyed and inspected lakes, harbours, reservoirs and stormwater assets for clients across New Zealand, from Auckland to the West Coast. A few highlights:

2020
Founded in the Waikato, and on the water ever since

26
Lakes mapped in a single programme for Waikato Regional Council

22.3 km
Round trip by our water-quality monitoring boat in a single mission

6
In-house engineers, developers & operators

WHERE WE WORK

Lakes / Harbours & coastlines / Drinking-water reservoirs / Stormwater ponds & wetlands / Settling & treatment ponds

WHO WE WORK FOR

Regional councils / District & city councils / Water utilities / Engineering contractors / Energy companies / Researchers

SHE Pre-Qual

Prequalified for safety, health & environment

Built in NZ

Our vessels and software are designed and built in-house

What we do

We build our own systems, so we can match the equipment to the job: on the water, under it, or in the air above it.

01

Autonomous systems

Custom USVs, ROVs, rovers and drones, tailored to the job.

02

Survey & mapping

Bathymetry, topography and topo-bathymetry to centimetre-level RTK.

03

Underwater inspection

Reservoirs, ponds, culverts and structures, with no draining and no divers.

04

Monitoring & sampling

Water quality profiling and multi-depth samples from autonomous boats.

05

IoT & data

Sensor networks, telemetry, live dashboards and alerts.

06

Software & AI

Mission control, data platforms and AI cameras at the edge.

Our fleet



Hydromap
Autonomous bathymetry USV



Hydrohub
Sampling, monitoring & bathymetry USV



SludgeScan
Dual-frequency & multibeam sediment USV



MicroScan
Compact sonar boat for tight, shallow sites



DeepView ROV
Underwater inspection with live video



Rover
Ground vehicle for confined spaces



AeroMap
RTK drone for 3D mapping



SkyDrop
Waterproof water-sampling drone

Survey and mapping



Stormwater pond survey: USV sonar, drone mapping and the finished 3D model.

Multibeam and single-beam sonar, RTK positioning and drone photogrammetry, combined into one accurate picture of the site, then processed into depth maps, elevation models and 3D outputs tailored to what you need to decide.

- Depth & contour maps
- Digital elevation models
- Integrated 3D models
- Sediment depth & volume
- Cross-sections
- GIS-ready data

Services: **Bathymetry** · **Topography** · **Stormwater ponds**

Inspection, monitoring and IoT



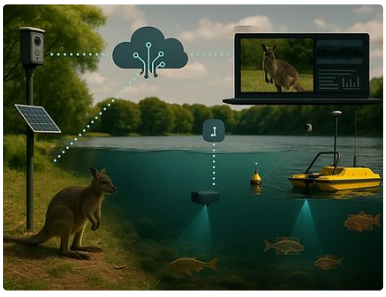
Underwater & confined-space inspection

Our ROV and rover inspect reservoirs, ponds, culverts, outlets and structures with live HD video. No draining, no divers and no confined-space entry.



Water-quality monitoring & sampling

Our sampling vessel profiles the water column with a precision winch: chlorophyll-a, phycocyanin, turbidity, DO, pH and temperature, plus 3 L samples from up to 15 m deep.



IoT & AI cameras

Solar-powered AI cameras detect pest animals such as wallabies at the edge, in remote places with little power or signal. Data flows to live dashboards and alerts.

More: **Reservoir inspection** · **Environmental monitoring** · **Solutions**

How we work

The same team handles your job from the first call to the final data. There are no hand-offs between a hardware supplier, a survey crew and a data processor.

01 **Scope**

We review the site with you, agree the outputs you need, and choose the platform and method that suit the site, not just the one we happen to own.

02 **Capture**

Our operators deploy the USV, ROV, rover or drone with RTK positioning, working to our SHE Pre-Qual safety system. No divers, no confined-space entry.

03 **Process**

Sonar, imagery and sensor data are processed in-house into depth maps, sediment volumes, cross-sections and 3D models.

04 **Deliver**

Clear reports, GIS-ready data and, where it helps, an interactive 3D portal. We show how confident we are in every figure.

Safer
Robots go where people shouldn't

In-house
Hardware, software & data under one roof

Fit for site
The method matched to each job

Honest data
Coverage and limits stated plainly

Safety, quality and biosecurity

Robots take people out of risky places. We still work to the standards councils and utilities expect of any contractor on their sites.

SHE Pre-Qual prequalified

Our safety, health and environment systems are independently prequalified.

No divers, no confined-space entry

ROVs, rovers and USVs go into the water and the tight spaces, so people don't have to.

Biosecurity between sites

IP67-rated, waterproof equipment with a rapid decontamination process between waterways.

Survey-grade positioning

RTK GPS for centimetre-level accuracy on every reading.

Honest about the data

Coverage, methods and limits are stated with every figure we deliver.

Built and supported in-house

We design, build, operate and maintain our own systems, so one team is accountable for all of it.

Case studies

A selection of recent projects for New Zealand councils, utilities and contractors.

These case studies are a selection of our work. Many other projects for clients across New Zealand aren't featured here.

01 **Western Bay of Plenty District Council**
11 stormwater ponds, Ōmokoroa · sonar, RTK & 3D portal

02 **Waikato Regional Council**
Water-quality monitoring & sampling in algal blooms · long-range USV

03 **Buller District Council**
Drinking-water reservoir, Westport · ROV & rover

04 **Auckland Council · Glasgow Contractors**
Hobsonville Point jetty · multibeam USV & RTK drone

05 **Waikato Regional Council**
26 lakes mapped · bathymetry programme

06 **Ōtorohanga District Council**
Aotea Harbour topo-bathymetry · settling ponds

07 **Lake Ngaroto, Waipā**
108 ha bathymetry & side-scan in 3 days

08 **Dutton Stormwater**
Underwater stormwater inspection by ROV

Eleven stormwater ponds, Ōmokoroa

The council needed to know how much water and sediment its Ōmokoroa stormwater ponds and wetlands hold, to plan and prioritise maintenance. Between June and August 2026 MĀKI surveyed eleven of them, matching the method to each site: multibeam sonar, sonar boat surveys, and RTK staff shots where a pond was too shallow or overgrown for a boat.

Each pond was delivered as a survey report plus a private online portal: a 3D model of ground, water and sediment, with a cross-section tool the council's engineers can use themselves. Every pond also shows how much of it lies close to a real measurement, so engineers know how far to trust each figure.

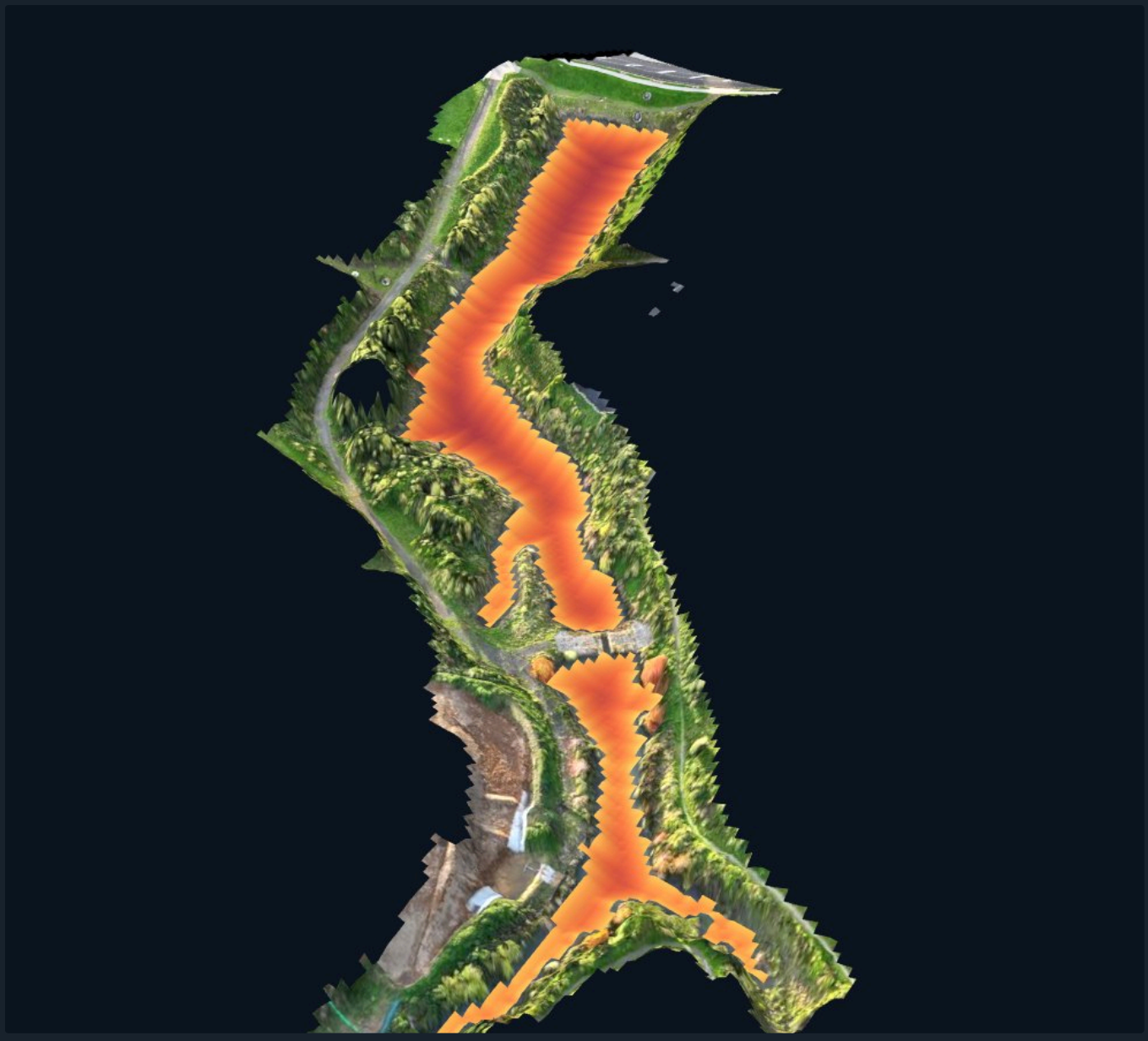


11
ponds & wetlands

3
survey methods

3D
portal with a cross-section tool

A stormwater pond in 3D



Ground
banks & surrounds, drone-mapped

Water
surface area & volume held

Sediment
depth, spread & volume

Confidence
coverage shown for every pond

Demo data: the pond bed and depths in this model are illustrative, not survey results. Clients receive full-resolution models and figures in their own private portal. **Read the case study →**



CASE STUDY · WAIKATO REGIONAL COUNCIL

Sampling algal blooms without putting people on the water

During a cyanobacteria bloom the water can be toxic. Sampling from a boat or jetty puts staff at risk, and a few fixed points say little about how the bloom is spread.

Since March 2024 we've worked with Waikato Regional Council's lake scientists to do this remotely. Our autonomous boat follows programmed routes across the whole lake, logging water quality as it goes, lowering its sensors to set depths and collecting samples of up to 3 litres by winch. The programme ran across several lakes over the 2024–25 summer.

0
staff on the water during a bloom

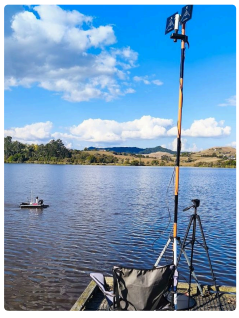
6
parameters, logged continuously

15 m
winch for readings and samples at depth

22.3 km
round trip in a single mission

Parameters: chlorophyll-a, phycocyanin (the cyanobacteria indicator), turbidity, dissolved oxygen, pH and temperature.





An antenna mast on the shore gives a direct radio link up to 2.5 km. With 4G and Starlink now on board, the boat stays connected well beyond that. Every reading is tagged with its RTK GPS position and streamed back live.

Long-distance monitoring, live data



The 22.3 km mission was a round trip: out to the turnaround and back to the same launch point. Map illustration based on the real route.

22.3 km
round trip in a single mission

4
sampling sites along the way

6
parameters, logged continuously

4G + Starlink
for live data well beyond radio range

Readings along the route ILLUSTRATIVE

Chlorophyll-a µg/L



Phycocyanin µg/L



Turbidity NTU



Dissolved oxygen mg/L



pH



Temperature °C



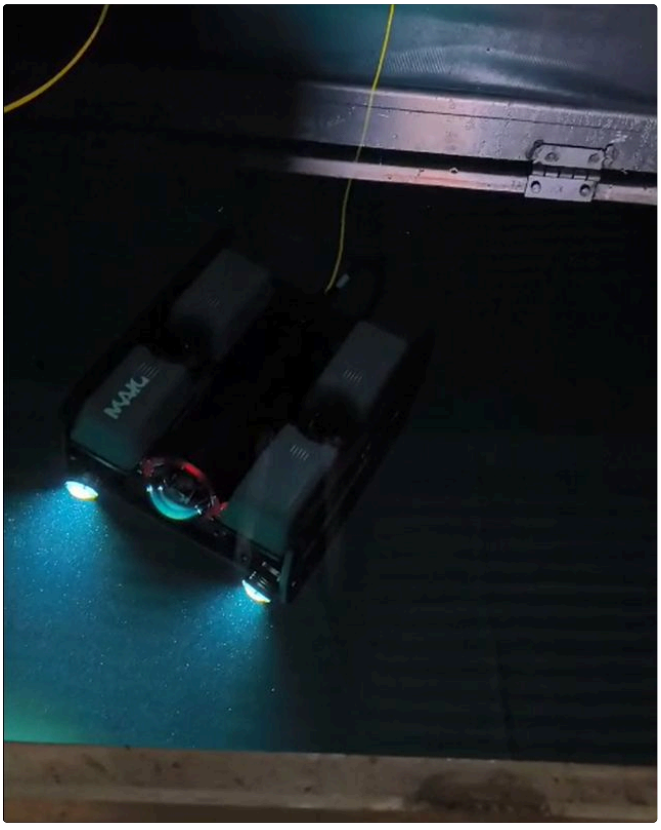
0 km 22.3 km | 0 km 22.3 km | 0 km 22.3 km

Read more: **Case study** · **Environmental monitoring & sampling**

Inside a drinking-water reservoir, Westport

Inspecting a drinking-water reservoir normally means taking it out of service: isolating it, draining it, gas-testing the air and sending someone into a confined space.

For Buller District Council, MĀKI inspected the reservoir in Westport with the water left in it. Our ROV surveyed the inside of the reservoir underwater, while our rover carried out the inspection on top of it. Together they gave the council a full visual record of the asset.



The ROV inside the reservoir



The rover, used on top of the reservoir

Watch the inspection



No draining

Inspected with the reservoir full

No confined-space entry

Nobody had to go inside

Inside and on top

ROV underwater, rover on the structure

Recorded footage

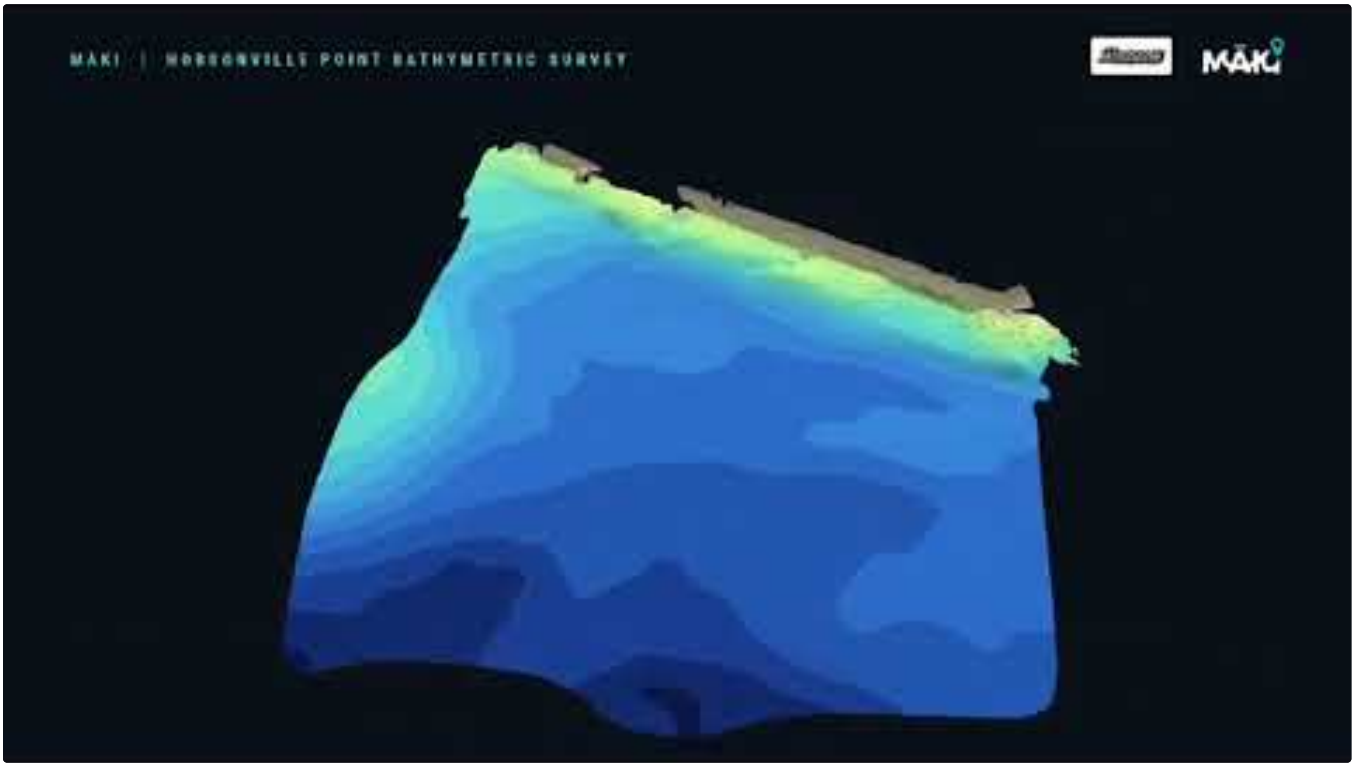
A visual record for condition reporting

Read more: **Full case study** · **Reservoir inspection service**

Under the jetty at Hobsonville Point

With Glasgow Contractors, on behalf of Auckland Council's Healthy Waters and Flood Resilience department, MĀKI completed a detailed bathymetric and topographic survey at Hobsonville Point on a tight timeframe.

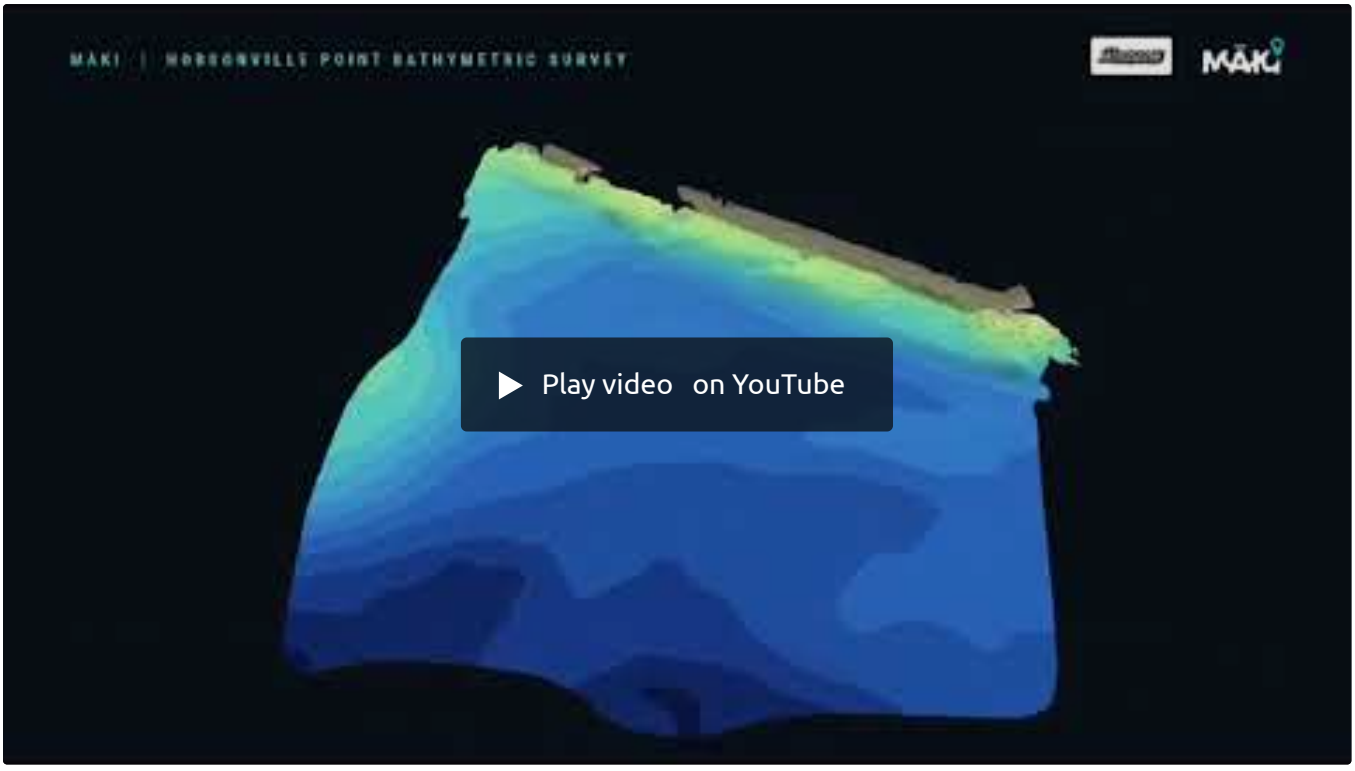
Our pilot drove MĀKI's in-house USV, fitted with multibeam sonar, underneath the jetty, a confined space that crewed survey boats struggle to reach. An RTK drone captured the exposed coastal structures and surrounding land, and the two datasets were combined into one integrated 3D model.



Multibeam seabed model, coloured by depth.

- ✓ Multibeam sonar USV
- ✓ RTK drone photogrammetry
- ✓ Under-jetty surveying
- ✓ Integrated 3D model

Watch the survey



Under the jetty

Data where crewed boats struggle to reach

Above + below

Sonar and drone data in one dataset

3D model

The structure and its surroundings, together

Fast

Mobilised quickly for a tight timeframe

Read more: **Full case study** · **Bathymetry service**

26 Waikato lakes mapped



Waikato Regional Council needed consistent depth data across many of its lakes to plan sediment work, track lake volumes and assess habitat. Crewed survey boats are costly and often impractical on small or sensitive lakes.

Our autonomous boats follow programmed survey lines across each lake, logging depth with single-beam and multibeam sonar positioned by RTK GPS. The data is processed into depth grids and 3D models in the council's GIS formats, giving a baseline that can be re-surveyed to show change over time.

26	RTK	GIS
lakes in one programme	positioning on every sounding	data in the council's own formats

Read more: **Case study** · **Bathymetry**

Harbour and settling ponds



Aotea Harbour topo-bathymetry

USV sonar below the water and drone photogrammetry above it, merged into one 3D dataset, including the geometry of the harbour's stone wall for engineering use.

Case study →



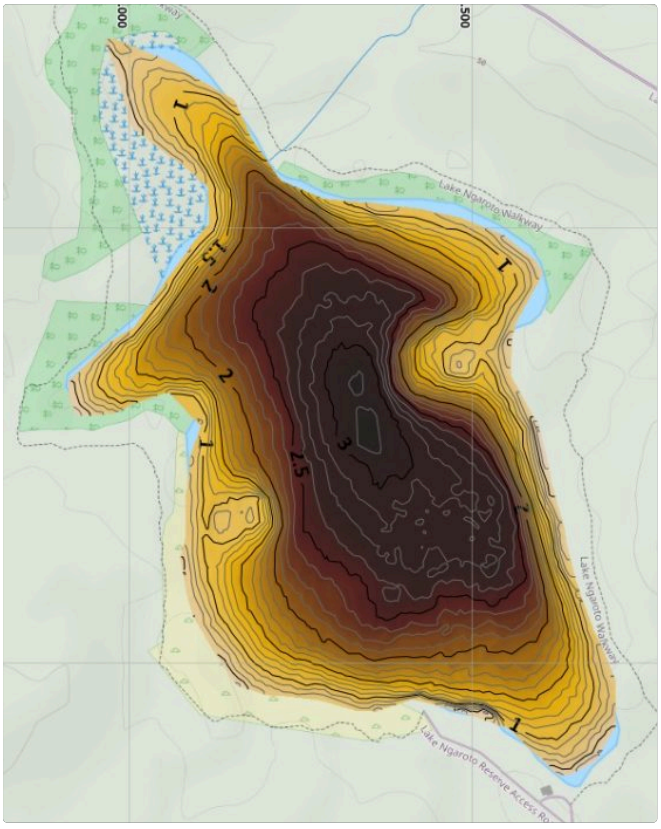
Settling-pond sediment volumes

Sediment scanning of settling ponds in Ōtorohanga town centre, giving the council sediment volumes and 3D models to plan maintenance before capacity runs out.

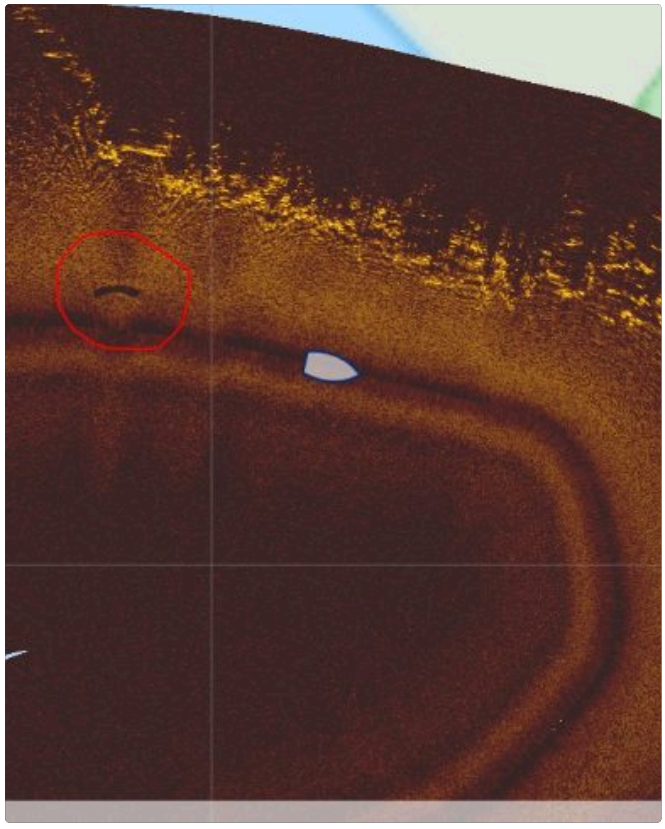
About settling-pond mapping →

108 hectares in three days

A full bathymetric and side-scan sonar survey of Lake Ngaroto from MĀKI's autonomous inflatable e-kayak, producing a complete depth map, a 3D lake model and a side-scan mosaic of the lake floor.



Depth map (contours in metres)



Side-scan sonar of the lake floor

108
hectares surveyed

3
days on site

55 m
unknown submerged structure found

Full case study →

Underwater stormwater inspection



Working alongside Dutton Stormwater, MĀKI used an ROV to inspect submerged stormwater infrastructure without anyone entering the water, and recorded footage for engineers to assess.

No divers
nobody enters the water

No draining
assets stay in service

On record
footage for engineers to review

The same approach suits ponds, culverts, outlets and other submerged structures.



INNOVATION

Research with councils and universities

We work with regional councils, universities and industry partners to develop and validate new technology for environmental monitoring and automation.

Through partnerships with organisations such as Waikato Regional Council and Auckland University of Technology, our engineering and software teams contribute to studies in autonomous sampling, IoT integration and AI-based sensing.

We welcome new research and development partnerships that combine scientific insight with engineering innovation.

News: **MĀKI at Fieldays 2025** · **Freshwater Conference 2025**

Development roadmap



3-in-1 integrated platform

A USV that acts as an "aircraft carrier" for a drone and an ROV. Solar and hybrid power keep all three charged for long missions; the drone returns to its dock to recharge, and the ROV deploys from the boat's central bay. It's all operated wirelessly, with no one on the vessel.



Submerged pipe inspection & 3D modelling

A robotic crawler for wet or submerged pipes, with LiDAR, sonar and HD cameras. SLAM positioning works without GPS to build 3D point clouds and digital twins for defect finding and asset management.



Amphibious USV

A survey vessel that drives itself from land into the water and back out again. It can launch where there is no ramp or jetty and cross between ponds and banks without being carried, opening up sites that are hard to reach today.

Clients and partners

Waikato Regional Council — Te Kaunihera ā Rohe o Waikato

Western Bay of Plenty District Council

Buller District Council — Te Kaunihera O Kawatiri

Auckland Council — Te Kaunihera o Tāmaki Makaurau

Watercare

Te Kaunihera ā-rohe o Ōtorohanga District Council

Mercury

Glasgow Contractors

Dutton Stormwater

Vernon Developments — Cartage, Earthmoving, Aggregate Recycling

Conhur — the new organic

Te Wānanga o Aotearoa

Submerged — Environmental & Scientific Dive Services

deepdive division

Torpedo7

PAKnSAVE

Contact us

Talk to us about a one-off survey, an ongoing monitoring programme or a custom system.

Get a quote

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